Agenda

Introduction

Functional statements

- Corporate Business Line
- **Operations Business Line**
- Projects & Engineering Business Line
- Security & Industrial Services Business Line
- Business Administration Business Line
- Human Resources Business Line
- Executive Director Office

GOV

Source: KJO, Arthur D. Little Analysis

Petroleum Engineering & Exploration Department (PED)

Petroleum Engineering & Exploration Department (1/5)

(All divisions should perform their role within the hierarchy of their department)

Department name	Petroleum Engineering & Exploration Department (PED)	**Reporting to**	EDO

Planning & Business Support Section

- **Manage the Petroleum Engineering & Exploration Department**
  - Develop, update, review and implement the policies, procedures and processes for PED
  - Coordinate, develop and administer all training for PED
  - Set the KPI targets for PED in coordination with CPD
  - Define remedial actions and develop action plan for underperforming KPIs
  - Measure and report on the performance of PED
  - Prepare regulatory reports
  - Design and implement improvement measures for PED
  - Develop the PED department business plan with input from divisions (5 and 20 years plans)
  - Manage the budget (including managing AFE and AFC)
  - Develop manpower plan in coordination with POD
- **Manage contracts**
  - Prepare SOW, terms and conditions for all departmental contracts in accordance with KJO processes
  - Develop purchase requisition
  - Monitor, evaluate, and report on contractor performance
  - Manage PED contracts and contractors (e.g change orders, variations etc.)
  - Confirm invoices meet contract terms and conditions
  - Support on contract negotiation, finalization, renewals, amendments, and renewals

Source: KJO, Arthur D. Little Analysis

■ Role ➢ *Responsibility*

Petroleum Engineering & Exploration Department (2/5)

(All divisions should perform their role within the hierarchy of their department)

Department name	Petroleum Engineering & Exploration Department (PED)	**Reporting to**	EDO

Geoscience Division

- **Provide geological and geophysics support to drilling activities**
  - Provide recommendations for well planning and workover plans
  - Monitor geo-steering activities during drilling operations
  - Monitor formation evaluation data acquisition activities
  - Prepare the geological well completion report
- **Provide geological and geophysics inputs for well planning**
  - Provide recommendations for reservoir characterization
  - Provide input to completion and workover plans
- **Manage geological and geophysics data**
  - Gather and store geological, seismic, and formation evaluation data
  - Check data for validity
- **Manage and plan coring operations and studies**
  - Plan and execute coring operation
  - Manage and supervise coring studies and analyses
  - Manage core house and cutting stocks for different studies requirements
- **Define exploration and appraisal program**
  - Develop exploration strategies
  - Plan seismic exploration surveys
  - Perform geological studies
  - Support well planning and drilling engineering studies
- **Conduct exploration and appraisal activities**
  - Conduct seismic exploration surveys
  - Collect and store core/cutting samples
  - Manage and gather formation evaluation data from exploration program
- **Evaluate exploration and appraisal wells**
  - Interpret seismic and formation evaluation data
  - Perform basin analysis and petrophysical activities
  - Identify future wells for appraisal
  - Handover successful exploration wells for appraisal

Source: KJO, Arthur D. Little Analysis

■ Role ➢ *Responsibility*

Petroleum Engineering & Exploration Department (3/5)

(All divisions should perform their role within the hierarchy of their department)

Department name	Petroleum Engineering & Exploration Department (PED)	**Reporting to**	EDO

Integrated Modelling & Studies Division

- **Appraise wells for economic viability**
  - Develop asset model and update as required
  - Estimate hydrocarbon potential
  - Build field development scenarios
  - Evaluate economic viability
  - Propose development plans
  - Handover to field development
- **Conduct reservoir characterization activities**
  - Process and analyze seismic characterization surveys
  - Build and update geological models
  - Manage the testing and data collection from core/cutting samples
  - Perform petrophysical activities, interpretation and reservoir characterization
- **Develop and maintain reservoir models**
  - Develop and maintain both static and dynamic reservoir models
  - Estimate reserve replacement ratio
  - Interface with reservoir management as required
- **Manage reservoir data**
  - Gather and store reservoir data, and formation evaluation data
  - Check data for validity
- **Conduct Pressure transient analysis (PTA)**
  - Conduct Pressure transient analysis
  - Define well testing build up/falloff program
  - Conduct well testing interpretation
  - Integrate well testing static model with simulation model
  - Perform well testing interference study
  - Evaluate well productivity and promote remedial job
  - Develop well behavior study with 3rd party
  - Define exploratory well testing (DST) objectives
  - Write exploratory well testing (DST) programs
  - Support well testing (DST) operations during execution
  - Provide all exploratory well testing (DST) requirements
  - Plan number of well tests (DST) test per well
  - Integrate well testing (DST) equipment with completion equipment
  - Witness well testing (DST) execution

Source: KJO, Arthur D. Little Analysis

■ Role ➢ *Responsibility*

Petroleum Engineering & Exploration Department (4/5)

(All divisions should perform their role within the hierarchy of their department)

Department name	Petroleum Engineering & Exploration Department (PED)	**Reporting to**	EDO

Data Management & Technology Division

- **Develop, implement and manage technologies**
  - Identify technological gaps and challenges
  - Define explore and appraise technological requirements
  - Manage the implementation of technologies
  - Manage, maintain, upgrade explore and appraise technologies
  - Define training needs and coordinate with TCD for specialized explore and appraise technology training
  - Provide support for technical issues
  - Scout for innovative technologies for continuous improvement
- **Manage data related to upstream Oil & Gas**
  - Gather, reformat, catalogue and index the documents and data.
  - Archive Technical reports and safeguard Seismic Tapes in the Data room
  - Transcribe Seismic SEG-Y tapes, copying and quality
  - Load and retrieve of seismic data into data management system
  - Load documents into E&P Physical Data Management System
  - Load, retrieve transfer all E&P data into the E&P Corporate Data management system
  - Load, retrieve transfer well log data into the E&P Well log database
  - Perform data validation
- **Develop, implement and manage E&P Systems**
  - Plan, adopt and manage E&P computing technology Systems and applications.
  - Develop and Manage E&P Corporate Data management system.
  - Perform Backup of all PED Data, Servers and Databases in regular basis
  - Operate & maintain the Visualization Center & Real Time Geosteering Center.
  - Manage servers and maintain the database
  - Upgrade PED specific software application and databases.
  - Update system specific policies and procedures.
- **Manage Geographic Information Systems (GIS) and cartography**
  - Generate Data through GIS
  - Convert data into maps.
  - Gather subsurface and surface maps for approved projects
  - Update GIS with the subsurface and surfaces maps

Source: KJO, Arthur D. Little Analysis

■ Role ➢ *Responsibility*

Petroleum Engineering & Exploration Department (5/5)

(All divisions should perform their role within the hierarchy of their department)

Department name	Petroleum Engineering & Exploration Department (PED)	**Reporting to**	EDO

Reservoir Management Division	
■ **Determine reservoir depletion strategies** ➢ Develop, update, and maintain depletion strategies ➢ Develop production plans and perform best sweep efficiency ➢ Conduct long term field life planning ■ **Manage production optimization** ➢ Monitor and model reservoir production ➢ Interface with production departments as required ➢ Advise on reservoir optimization initiatives ➢ Identify and recommend technologies for implementation ➢ Monitor water encroachment behavior ➢ Plan and analyze the cased hole logs ■ **Mange field life extension** ➢ Advise on well operating parameters ➢ Perform well placement strategy for water injection and oil producer ➢ Recommend and advise on well workover and well services programs ➢ Recommend and advise on field development programs ■ **Manage petroleum and reservoir data** ➢ Gather and store petroleum, and reservoir data ➢ Check data for validity	

■ Role ➢ *Responsibility*

Source: KJO, Arthur D. Little Analysis